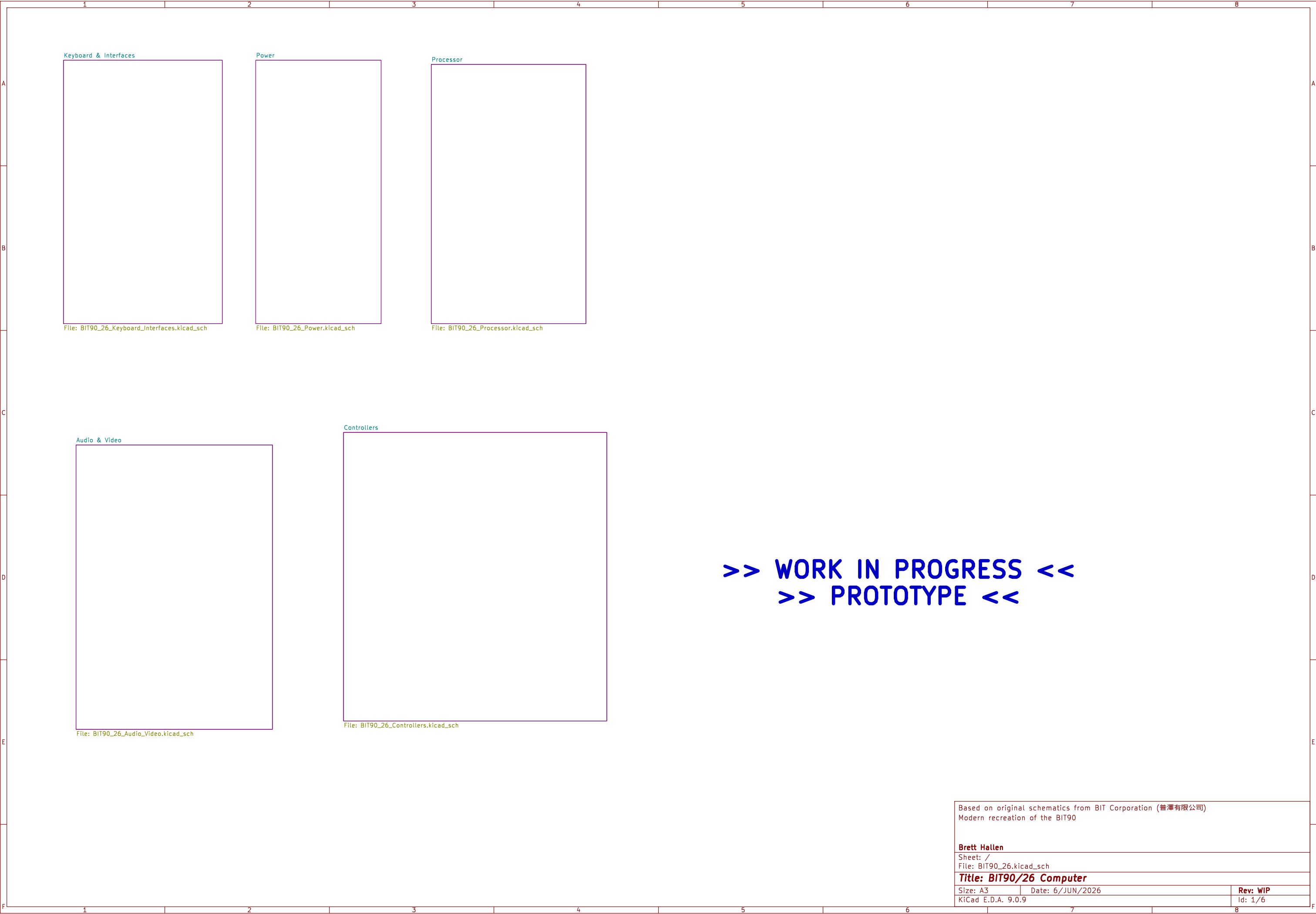
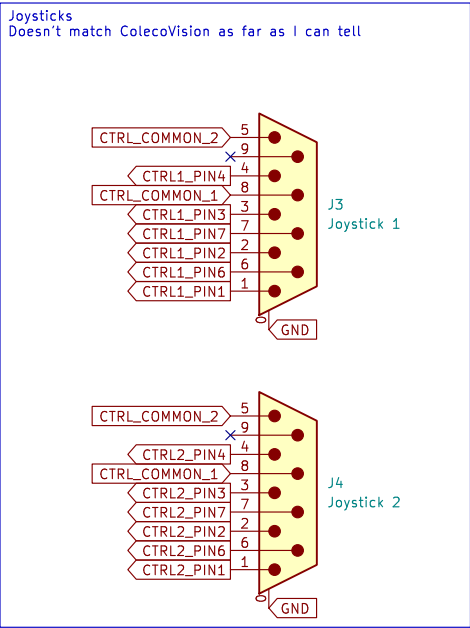
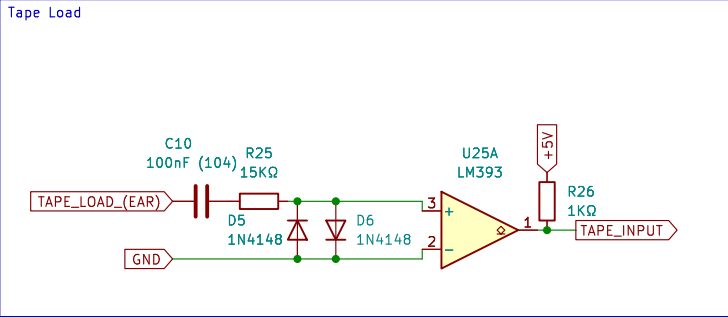
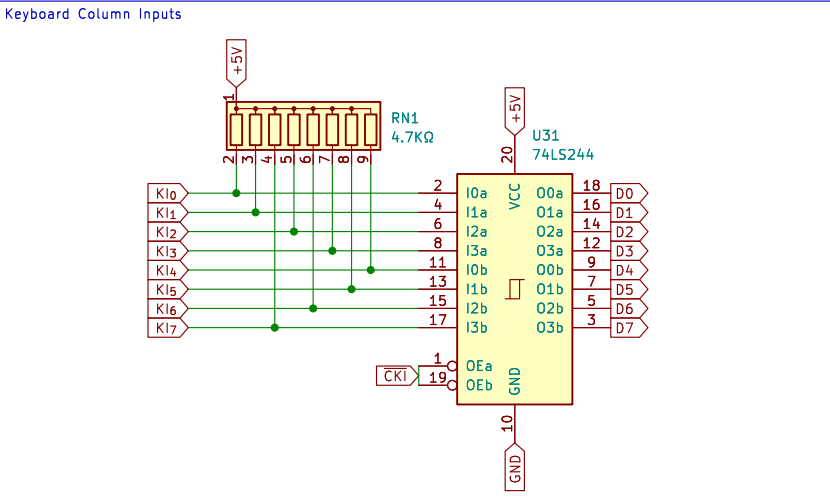
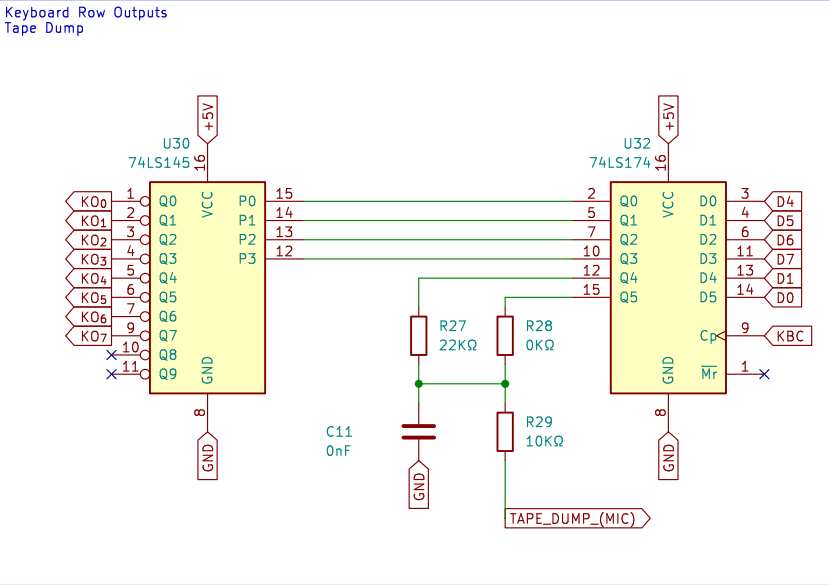
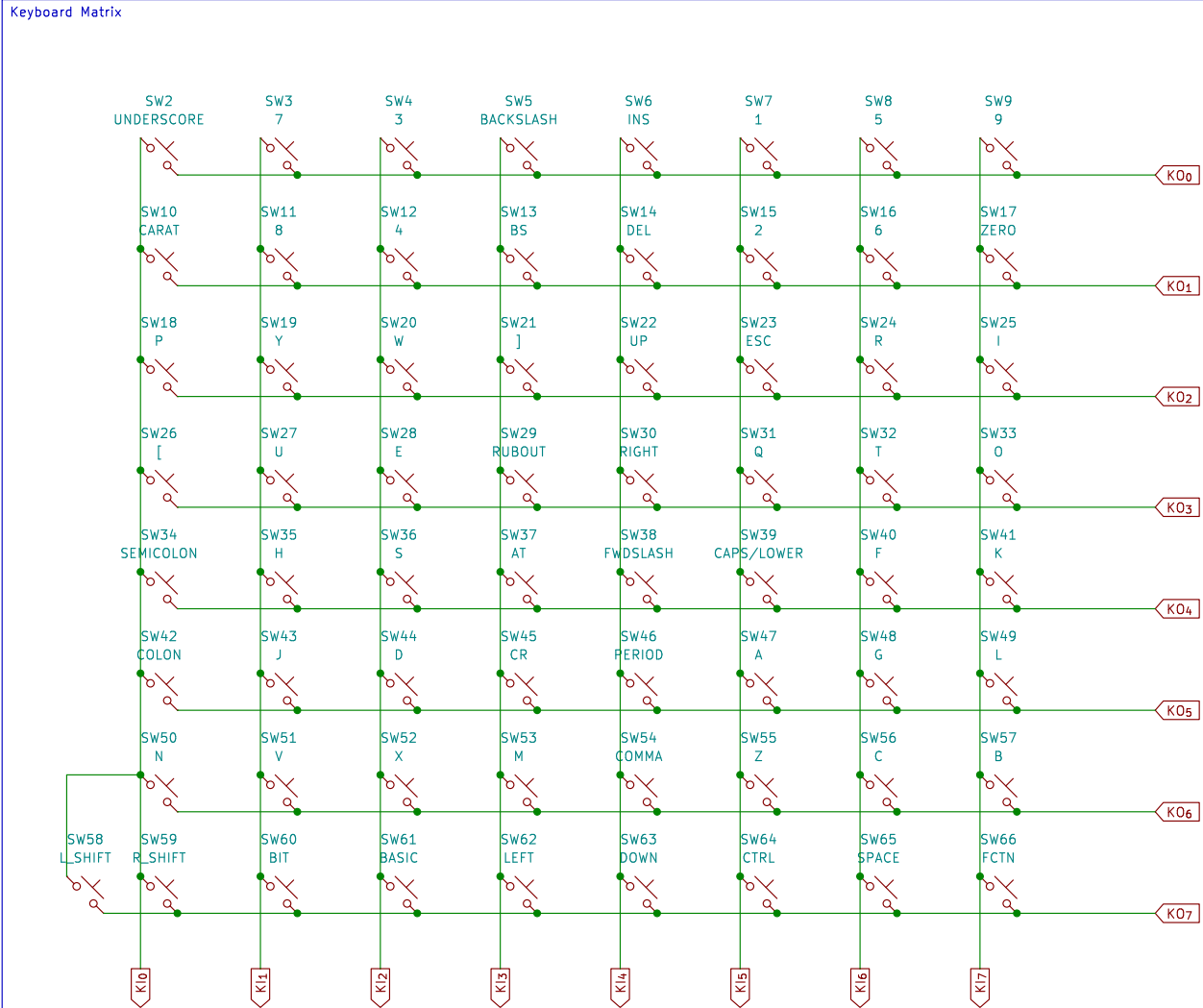




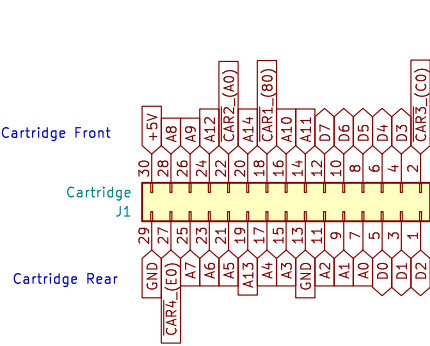
Based on original schematics from BIT Corporation (普澤有限公司) Modern recreation of the BIT90		
Brett Hallen		
Sheet: / File: BIT90_26.kicad_sch		
Title: BIT90/26 Computer		
Size: A3	Date: 6/JUN/2026	Rev: WIP
KiCad E.D.A. 9.0.9	Id: 1/6	



System Expansion (Edge Connector)

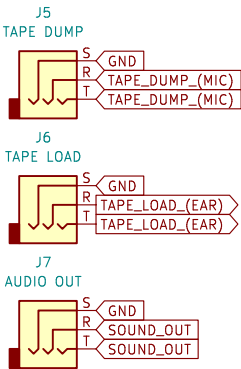


ColecoVision Cartridge

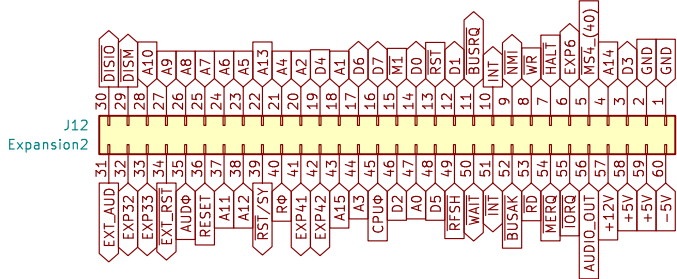


Tape & Audio
On BIT90 these are combined in a DE9 plug along with composite video

Audio Output: 1.5Vp-p
Tape Load (EAR): 500mVp-p
Tape Dump (MIC): 90mVp-p
Video Output: Dedicated HDMI board (Pico9918)

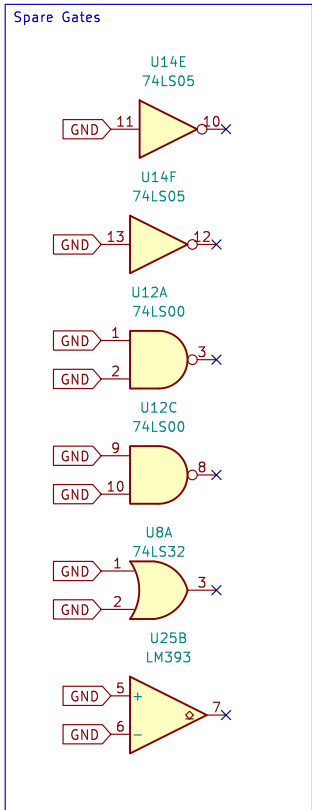
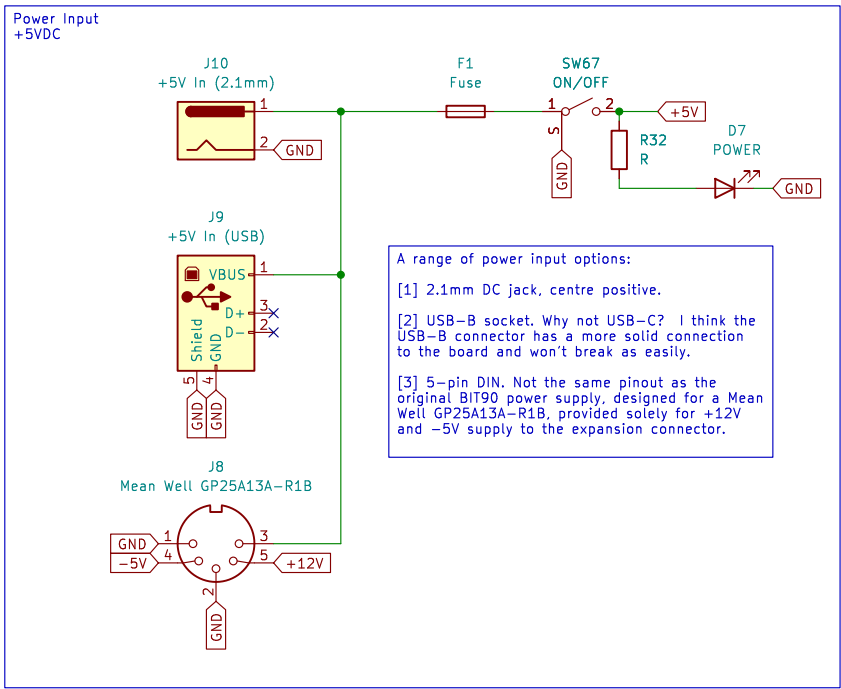
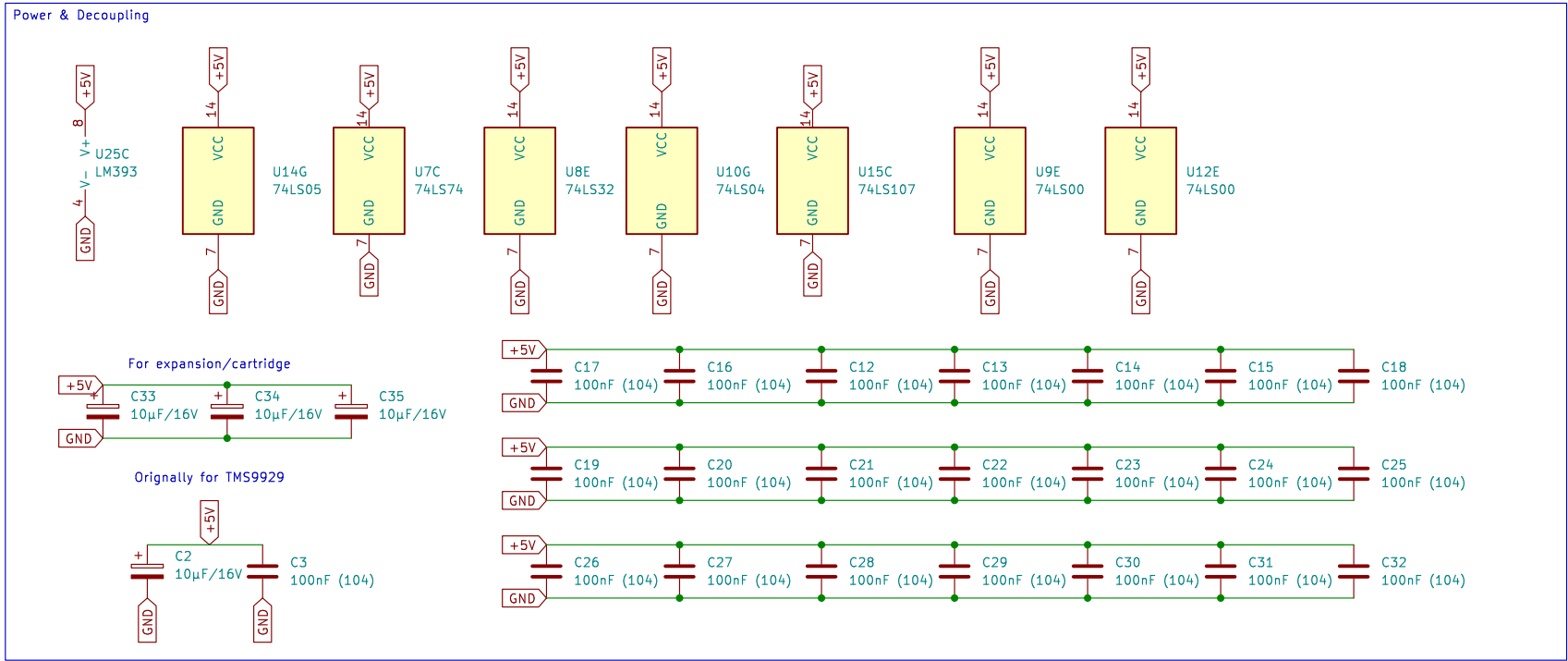


System Expansion (Pen Header)
Additional option for connecting/accessing the expansion bus



These expansion bus pins aren't connected anywhere else at the moment ... remove?
EXTVE (pin 32)
EXTV (pin 33)
BUSRQ wasn't connected to Z80, just pulled high, so I have connected it through.
INT wasn't connected either, just INT via an inverter so I have connected it through.

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Modern recreation of the BIT90



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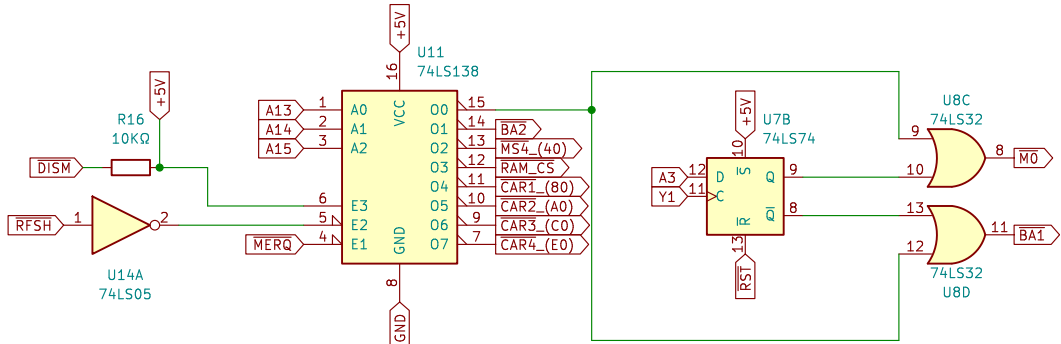
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File: BIT90_26_Power.kicad_sch

Title: **BIT90/26 Computer – Power**

Size: A3 Date: 6/JUN/2026
KiCad E.D.A. 9.0.9

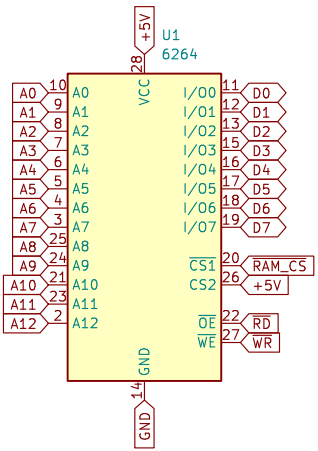
Rev: WIP
Id: 3/6

Chip Select

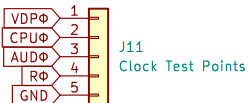


M0	ColecoVision BIOS
BA1 & BA2	BASIC
RAM_CS	System RAM
CAR1 to CAR4	Cartridge
MS4	Expansion

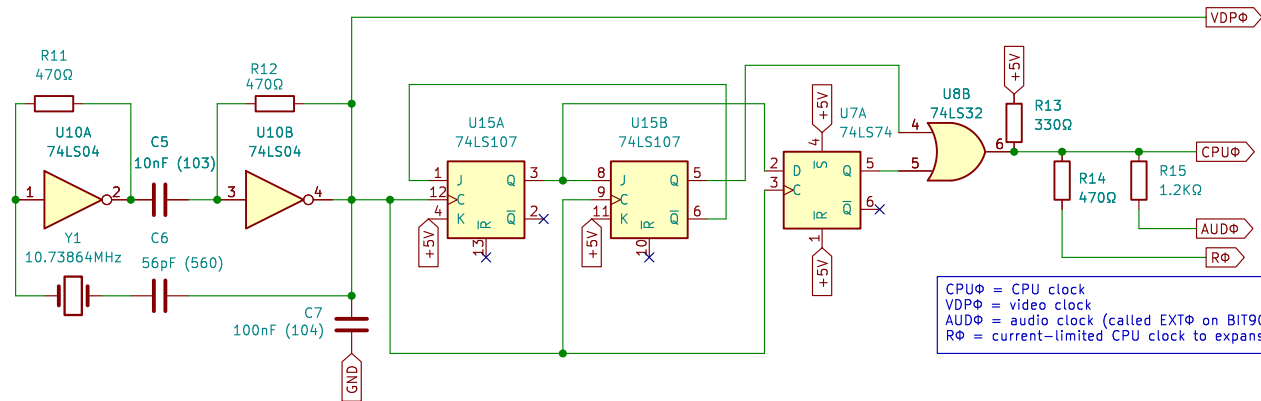
System RAM
BIT90 had 2KB originally in an 8KB block
Let's upgrade to 8KB
Upgrading further to 32KB will require additional logic
Let's see if this works first ...



Clock Test Points

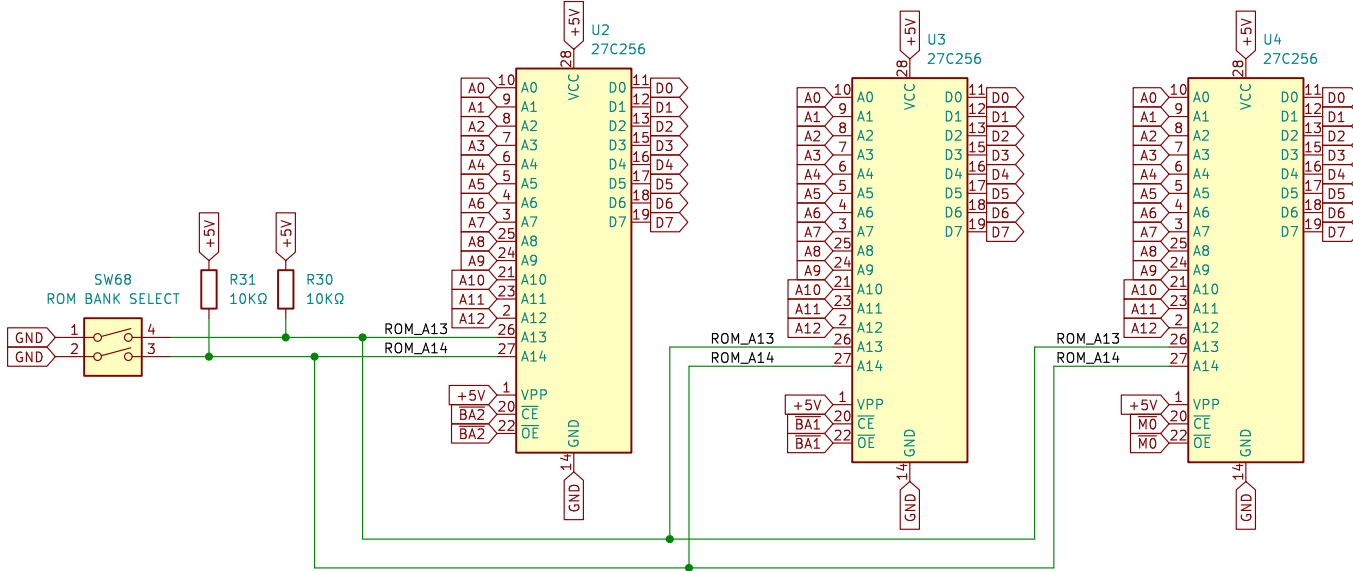


System Clocks



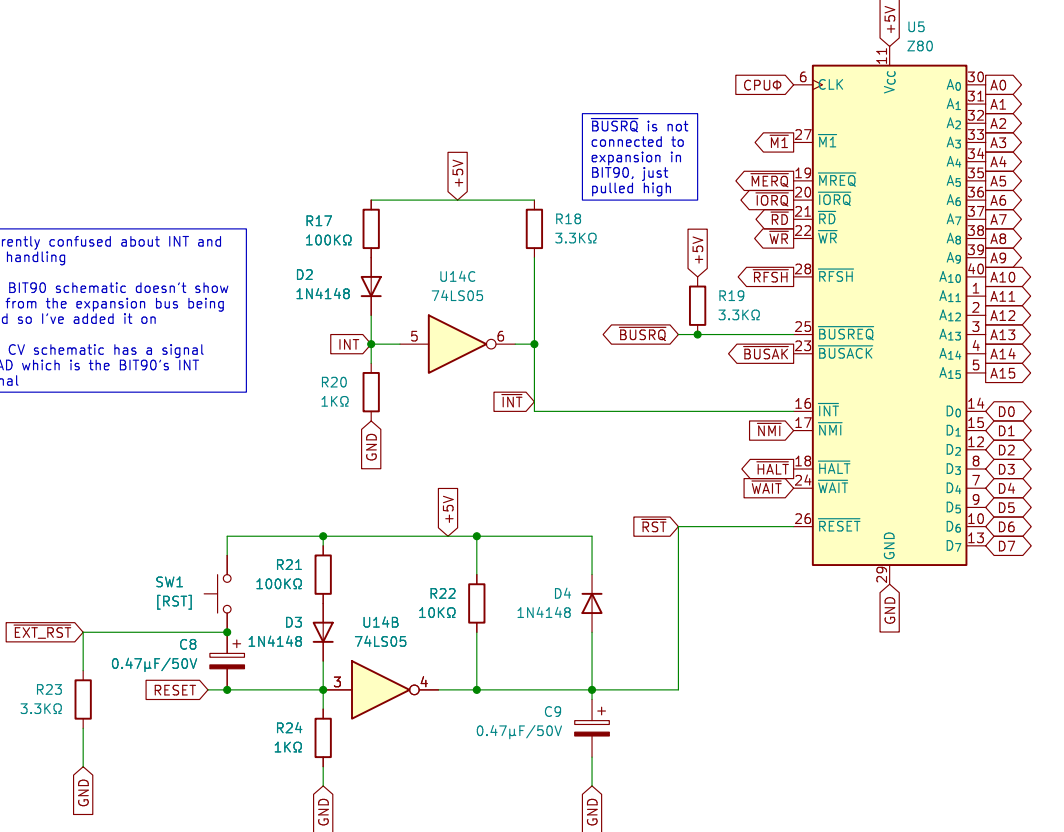
CPUΦ = CPU clock
VDPΦ = video clock
AUDΦ = audio clock (called EXTΦ on BIT90)
RΦ = current-limited CPU clock to expansion

System ROMs
BASIC1, BASIC2, ColecoVision BIOS
Original ROMs are 8KB
Using 27C256 allows four banks that can be switched



CPU & Reset Circuit

Currently confused about INT and INT handling
The BIT90 schematic doesn't show INT from the expansion bus being used so I've added it on
The CV schematic has a signal QUAD which is the BIT90's INT signal



Based on original schematics from BIT Corporation (普澤有限公司)
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Brett Hallen
Sheet: /Processor/
File: BIT90_26_Processor.kicad_sch
Title: BIT90/26 Computer – Processor
Size: A3 Date: 6/JUN/2026 Rev: WIP
KiCad E.D.A. 9.0.9 Id: 4/6

